

Attention Based Adaptive Graph Convolution Network for Weakly Supervised Video Anomaly Detection

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ABSTRACT

Weakly supervised anomaly detection in surveillance videos presents a significant challenge due to the difficulty of learning from video-level labels and the diversity of anomalous events. A detection method is presented in this work by constructing Feature similarity (spatial) and temporal consistency graphs and train an Attention based Graph convolutional network model along with residual layers to detect anomalies based on anomaly scores assigned and plot a anomaly plot with elevation points showing the presence of anomaly in the videos. By adding the attention and residual layers we are able to ensure that Attention dynamically focuses on anomalous nodes, while the residual layer ensures these critical features are not being lost in deep layers. For elaborate experiments using the UCF Crime dataset along with the presence of Attention and Residual layers along with our layered GCN model we achieved a AUC value of 92.15 %. Finally testing the proposed system with a raw Anomaly video we were able to produce accurate elevation points in the plot at the time of occurrence of anomalous events in the weakly supervised surveillance footages

1. Introduction

Video anomaly detection has become an increasingly vital area of research in modern surveillance, aiming to automatically identify deviations from normal behavior in video streams to enable timely responses to potential security threats [1]. However, achieving robust and accurate anomaly detection remains challenging due to factors such as variations in lighting, occlusions, diverse camera angles, and the inherent difficulty in defining 'normal' behavior across varied and unpredictable surveillance environments [2]. Traditional anomaly detection methods, when evaluated on many existing anomaly detection datasets [3], often rely on manually engineered features and shallow learning models, which frequently struggle to capture the intricate spatio-temporal relationships present in video sequences.

To address these limitations, deep and multiple instance learning techniques have emerged as promising alternatives [4, 5], leveraging their ability to automatically learn complex spatio-temporal features for enhanced anomaly identification. Among these, Graph Convolutional Networks (GCNs) have garnered significant attention due to their capacity to model structured relationships within video data, representing interactions and contextual dependencies to detect deviations from learned patterns of typical behavior [4]. While GCNs offer a structured way to represent relationships between entities (where nodes represent objects and edges represent spatial and temporal connections [6]), traditional GCNs often operate with static graph topologies, which may not optimally capture the dynamic nature of real-world video data.

This research introduces a novel video anomaly detection method that utilizes Attention based Graph Convolutional Networks to learn more effective graph representations for anomaly detection. In contrast to conventional GCNs with fixed graph structures, Adaptive GCNs can dynamically learn the optimal graph topology directly from the video data [7]. This adaptability allows the model to better discern between normal and anomalous events by more accurately capturing the evolving relationships between objects or regions of interest [8, 9]. Methods employing Spatial Temporal Graph Convolutional Networks with spatial and channel attention [10] and Temporal Graph Convolutional Networks with attention [11] have demonstrated the benefits of dynamic graph representations. Our approach builds upon these insights by incorporating adaptive mechanisms to learn crucial relational information. To achieve this, video sequences are first segmented into smaller units for localized analysis. Subsequently, discriminative features capturing both the appearance and motion information are extracted using a deep convolutional neural network specifically designed for spatio-temporal feature extraction, such as the Inception Inflated 3D (I3D) network resNet50 backbone [12]. The AGCN is then applied to model the relationships between these features within the dynamically learned graph structure, enabling more precise and context-aware anomaly detection. By adaptively refining the graph connections during training, and potentially integrating an attention mechanism for focused feature selection [11], the model can concentrate on the most salient interactions within a scene, thereby significantly improving anomaly detection performance. This paper focuses on the application of the proposed Attention based AGCN method for identifying anomalous activities in video data.

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2. Related literature study

Early research in video anomaly detection explored graph-based approaches to model relationships in video data. One notable method, the "Normal Graph," uses Spatial-Temporal Graph Convolutional Networks (ST-GCNs) to learn typical motion patterns from skeleton data [4]. By analyzing the spatio-temporal dynamics of human movement, it captures how body joints interact over time. The model predicts future motion sequences under normal conditions, flagging anomalies as significant deviations from these predictions. This work highlights the potential of graph-based learning for understanding normal human motion and detecting irregularities.

However, a challenge in weakly supervised video anomaly detection is bias in Multiple Instance Learning (MIL). MIL treats videos as "bags" of segments ("instances"), but it often focuses too much on specific instances, skewing results. To address this, researchers proposed an "unbiased" MIL approach that reduces selection bias, ensuring fairer learning from weakly labeled data [5]. Graph-based modeling was further refined by representing video scenes as graphs, where nodes are objects or regions, and edges capture their interactions [6]. Normal scenarios show consistent relationship patterns, while anomalies disrupt these structures. This approach effectively models spatial dependencies in surveillance environments.

To improve video representation, Adaptive Graph Convolutional Networks (AGCNs) were introduced [7]. Unlike traditional GCNs with fixed structures, AGCNs dynamically adjust graph topologies based on input data, enhancing flexibility in capturing spatio-temporal relationships. Another advancement, the "Anomaly Graph," uses dynamic GCNs to model evolving relationships in video data [8]. By tracking how interactions change over time, it identifies deviations from typical patterns in real-world scenarios.

For the case of multi-scale analysis, the Hierarchical Spatio-Temporal Graph Convolutional Neural Network (HST-GCN) was developed [9]. Its hierarchical structure examines data at different granularities, detecting anomalies that may appear as local irregularities, global inconsistencies, or temporal deviations. To improve ST-GCNs for human behavior recognition, researchers added spatial and channel attention mechanisms [10].

These help overcome challenges like environmental noise and viewpoint changes, making abnormal behavior detection more robust. For dynamic graphs (where connections change over time), AddGraph was introduced [11]. It uses attention-based Temporal GCNs to detect anomalies in evolving node and edge relationships. In order to aid these graph-based methods, deep learning models like ResNet50 and Inflated 3D Convolutional Networks (I3D) have been combined to extract both spatial and temporal features [12]. ResNet50 captures visual structures, while I3D extends into the temporal domain to model motion dynamics.

Another technique [13] tackles imbalanced data and complex feature extraction by using parallel deep learning architectures with attention mechanisms. This enhances

spatio-temporal feature learning for more accurate anomaly detection. Attention mechanisms also play a key role in the Video Anomaly Behavior Detection Method (AEG-NF) [14]. It combines spatial and channel attention graph convolution to capture global features and integrates them into a spatio-temporal network. Similarly, another study [15] uses spatial and channel attention graph convolution to build an attention-enhanced GCN, paired with normalizing flows to further refine anomaly scoring.

For adapting models to new surveillance scenarios with limited data, the Segmentation and Optical Flow Fine-tuning (SOFF) framework was proposed [16]. It fine-tunes models using segmentation and optical flow estimates via a Segmentation and Flow Output Network (SFO-Net). Lastly, a weakly supervised method incorporates clustering during training to better separate normal and abnormal events [17]. Since weak labels lack segment-level details, clustering helps the model learn clearer distinctions.

3. Proposed video anomaly detection pipeline

3.1. Pre-processing

3.1.1. Video Segmentation

The videos available in the UCF crime dataset have varying lengths in range of seconds, minutes and hours. So to keep things simple and useful we segment these videos on the basis of 16 frames per segment so that each video is divided into T number of segments depending upon the length of the video as shown in figure 1. By doing this segmentation process we have now divided each video into different segments which are collectively used for feature extraction in the next step.

3.1.2. I3D feature extraction

I3D (Inflated 3D ConvNet) is a pre-trained deep learning model used for video action recognition and spatiotemporal feature extraction. It extends 2D convolutional networks (like Inception-v1) into 3D to capture motion and temporal patterns in videos. I3D is widely used in action recognition and was primarily trained on the kinetics dataset. The key advantage of using this I3D feature extraction model in our research is to make use of the transfer learning, since this I3D model uses the pre-trained weights from large datasets like Kinetics-400 improving performance on smaller datasets and the 3D convolutions efficiently model temporal dynamics, outperforming traditional 2D CNNs.

For each video with T segments which was divided earlier we use the mix-5c layer of the I3D extraction to extract 2048 features. Hence for each video with T segments we get 2048-D features extracted and saved. Since T (number of segments of a video) is varying, to process this T, D further we need to have a fixed T . Hence for simplicity we fix $T=150$ as a standard value so that values below 150 are zero padded and above 150 are truncated. Additionally, I3D supports two-stream architectures (RGB and optical flow), but here we consider only the RGB features for our feature extraction. Now finally the output of this feature extraction step will

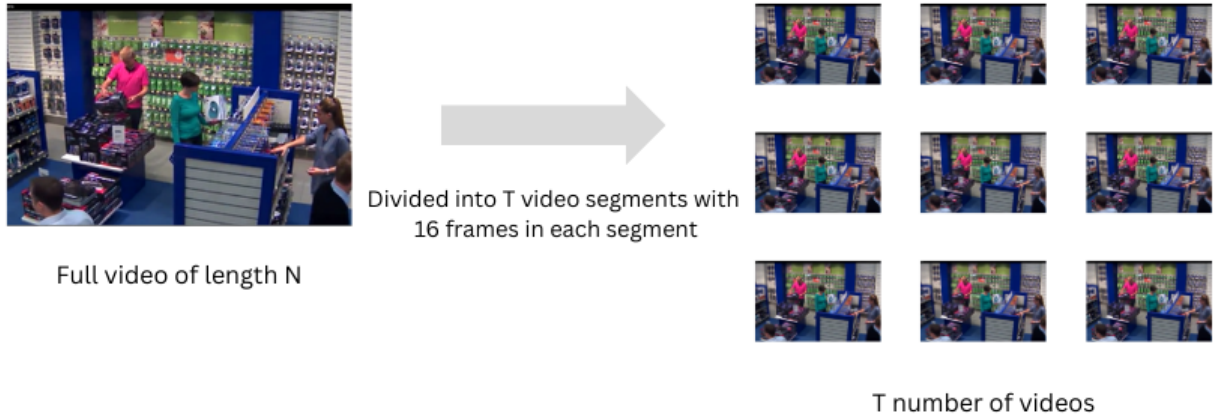


Figure 1: Process of 16 frames Video Segmentation

have a numpy file with an array of dimension (T,D) where T=150 and D=2048

3.1.3. Feature Size Reduction

Just before the next Graph construction step in order to reduce more complexity in the construction step we reduce the number of features D from 2048 to 512. We initialize target dimension as D=512 and use a linear fully connected layer to reduce D from 2048 to 512 and uses weights as an identity matrix and biases as zeros. This identity initialization helps maintain feature semantics during reduction.

3.2. Workflow

The Workflow of the presented anomaly detection system is shown in the figure 2. Once these features are extracted using the I3D model for each segments combined to a video and reduced in dimension the next step is to construct adaptive adjacency graphs for these extracted features in order to be loaded to the Graph Convolutional network to learn. Two graphs namely Feature similarity and Temporal consistency graphs which capture the feature and positional similarity arguments among the segments of a video are constructed based on graph learning mechanisms.

Post this graph construction now we need to construct a Graph Convolutional network model to understand and learn from the adjacency graph created. This model is constructed with necessary layers that can handle the given number of features and the context that holds and trained with the train data and fixed training parameters. Finally with our test dataset we test the model that was trained earlier using the model weights and get the Area under curve (AUC) score for the given test data.

3.3. Feature Similarity Graph AF

We have constructed our graphs from the equations proposed in the work [7] for both AF and AT given in the following sections

In this method we construct Feature similarity graph AF using the graph learning layer to model the relationship between the segments of the video, The relationship between segments here represents both the similarity and adjacency of them. This allows the graph structure to refine itself based on this data patterns, enabling more flexible and context aware modeling of segment interactions compared to direct predefined graphs. The dynamic adjustment of edge weights ensures the adaptability and thus improving the ability of the model to learn complex situations in the given video as given in the Eq 1 which was considered from the construction method given in [7]

$$A^F = \text{softmax}(\text{relu}(XW_1) \text{relu}(W_2^T X^T)) \quad (1)$$

3.4. Temporal Consistency Graph AT

To effectively model the sequential nature of videos, a temporal consistency graph AT is given such that it captures the inherent chronological relationships between video segments. Unlike manually constructed graphs, AT leverages the natural temporal ordering of frames to represent the flow and evolution of events over time. By preserving the original temporal dependencies, AT provides a robust foundation for analyzing dynamic video content while respecting its inherent sequential structure as given in the Eq 2 which was considered from the construction method given in [7]

$$A_{ij}^T = \exp(-|i - j|) \quad (2)$$

3.5. Global Graph A

This feature similarity and temporal Consistency graphs are combined to form the Global graph as shown in the Fig 3 just to make sure both features are inclusive to be trained by the model to extract better results. While AT enforces sequential structure, AF dynamically adjusts connections

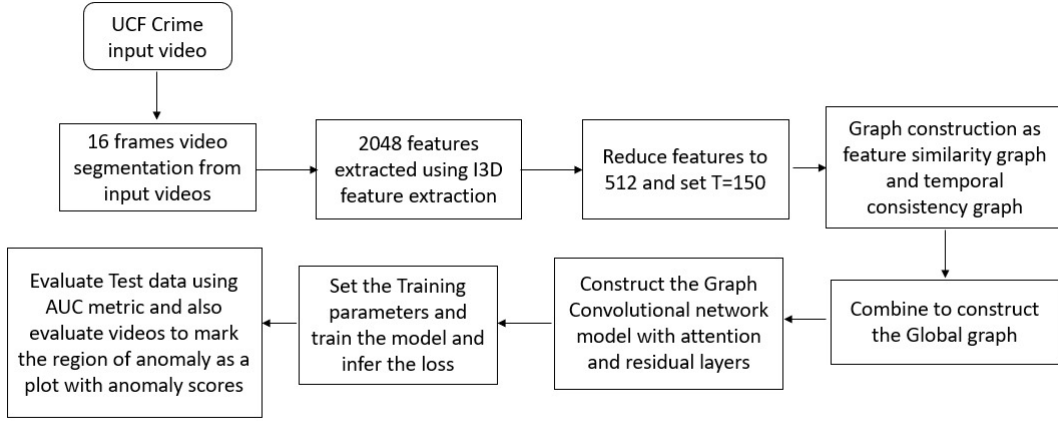


Figure 2: Our Proposed Workflow for Weakly Supervised Video Anomaly Detection.

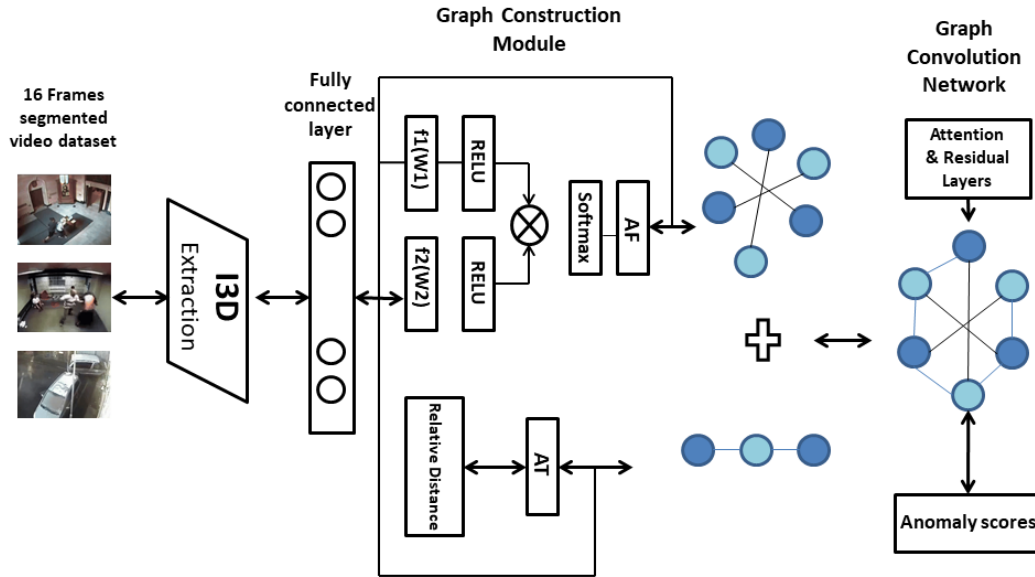


Figure 3: Our Proposed Architecture for Weakly Supervised Video Anomaly Detection

based on content similarity, allowing the model to detect non-linear or long-range dependencies and discrepancies between feature similarity and temporal consistency can signal anomalies more effectively.

3.6. Graph Convolutional Network Model

The GCN architecture processes video frames as nodes in a graph, where edges encode spatiotemporal relationships between regions. The fundamental unit of the model is the GCNLayer, which takes an input feature matrix X and an adjacency matrix A to perform graph-based message passing, where the operation follows as given in the Eq 3. The sequentially stacked multiple GCN layers are followed by Layer Normalization and ReLU activation. However, standard GCN layers suffer from limitations such as vanishing gradients and over-smoothing when stacking multiple layers. To address these, the model integrates three key

enhancements: residual connections, attention-based edge weighting, and skip connections as shown in Fig 4. The Base operation propagates features using adjacency matrix A and node features x . Here, our model consists of 3 GCN layers where the input feature dimensions given for the model is $(T, 512)$ where $T = 150$ based on our base paper with hidden dimensions assigned as $[128, 32, 1]$. The final output layer utilizes a sigmoid activation to predict anomaly scores. Dropout is employed to prevent overfitting.

$$X^{(l)} = \text{ReLU}(\text{Dropout}(AX^{(l-1)}W^{(l)} + R^{(l)}) \odot f_{\text{attn}}^l(A)) \quad (3)$$

3.6.1. Attention Mechanism

The attention mechanism introduces a learnable scaling factor to dynamically adjust the contribution of adjacency-based feature propagation. Unlike static graph convolutions,

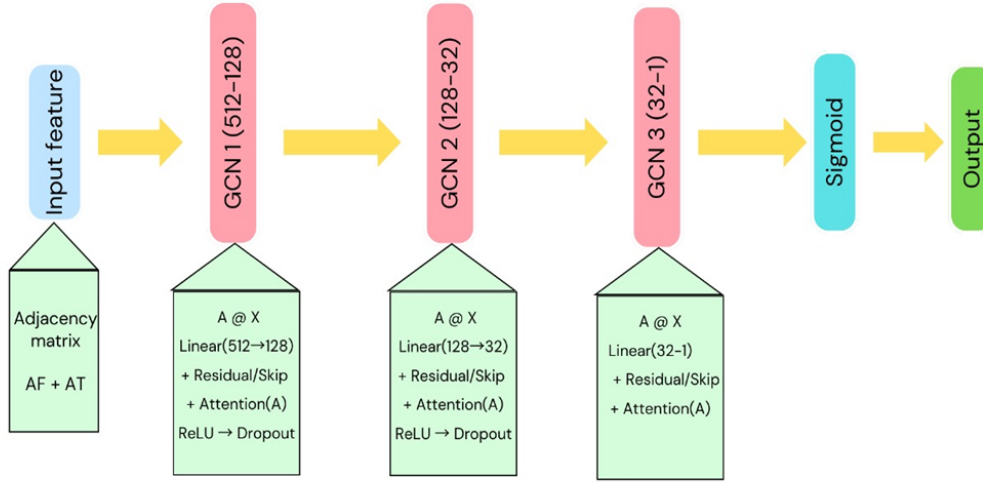


Figure 4: Proposed Attention Based GCN Model Architecture for Weakly Supervised Video Anomaly Detection

our model learns the importance of connections adaptively, leading to better representation learning: where it is initialized as 1.0 and updated during training. In our implementation, the self.attention parameter in the GCNLayer class allows the model to re-weight the adjacency matrix dynamically. This ensures that edges contributing more to anomaly detection receive higher influence, making the model more flexible and robust to noisy connections.

3.6.2. Skip Connections

Skip connections play a crucial role in stabilizing deep GCNs. In our model, skip connections transform the input features and add them to the output of the layer. This prevents information loss and enables better feature reuse across layers: where is a transformation matrix mapping input features to output dimensions. From our code, the self.skip-connection and self.skip-weight ensure that the original input information is preserved and added proportionally to the transformed features. The ability to skip transformations prevents excessive degradation of useful signal, particularly in deeper GCNs.

3.6.3. Residual Learning

Residual learning further enhances model depth by enabling gradient flow through identity mappings. This improves optimization and prevents vanishing gradients: where is the transformation applied by the GCN layer. In the code, the self.residual-transform ensures that even when input and output dimensions differ, a transformation aligns them before addition. This guarantees that deep networks do not suffer from learning instability or information loss.

3.6.4. Layer Normalization & Dropout

Normalizes activations within each layer to stabilize training and improve convergence speed. This is particularly beneficial in deep GCN architectures, where feature distributions can shift across layers. Randomly drops units during training, preventing over-reliance on specific connections and improving generalization. A dropout rate of 0.6 is used to effectively regularize the network.

3.6.5. K-max loss

K-max loss is designed to selectively focus on the top-K most informative predictions from a sequence, ensuring robust learning by emphasizing the most confident samples. Given predicted scores S and corresponding ground truth labels Y , the loss dynamically selects a subset of values for optimization. The top K highest scoring predictions are identified, followed by binary cross-entropy (BCE) loss being applied to the selected scores and their corresponding ground truth labels. The loss is computed for each sequence and averaged across the batch for stable optimization. The equations for this K-max loss is given in Eq 5 and Eq 4.

$$L_{k-\max} = \frac{1}{B} \sum_{b=1}^B \frac{1}{k_b} \sum_{i \in S_b} [y_i \log s_i + (1 - y_i) \log(1 - s_i)] \quad (4)$$

$$k = \max(1, \lfloor T \times 0.125 \rfloor) \quad (5)$$

- B - Batch size

- T - Number of segments
- s_i - Predicted anomaly score
- y_i - Pseudo-label
- S_b - Indices of top- k_b segments

4. Experimentation

4.1. Dataset

To detect anomalies from surveillance videos we use the UCF crime dataset introduced by the University of Central Florida (UCF) and contains real time surveillance videos depicting both normal and anomalous events. The dataset contains 1,900+ long untrimmed surveillance videos with 13 types of anomalous classes (Burglary, Shooting, Abuse, Shoplifting, Arson, Road Accident, Stealing, Explosion, Arrest, Robbery, Assault, Vandalism, Fighting) along with Normal category videos. We have used all these categories for our model training and experimentation purpose.

4.2. Training

The Labeling of data is done only on the video level as already mentioned before. The model is trained for 13 classes where each video is categorized as either normal or anomalous based on predefined event types. Categories such as Burglary, Robbery, Shoplifting, and Arson are labeled as anomalous (1), while the Normal category is labeled as non-anomalous (0). Labels are assigned at the frame level, with all frames in an anomalous video receiving a label of 1, ensuring frame-wise supervision. The model is trained for 100 epochs using the Adam optimizer (learning rate = 0.001, weight decay = 0.0005) with a batch size of 16 as shown in the Table 1. The dataset used for training consists of 701 anomaly videos and 705 normal videos. During the training the best and last weights are saved for testing purposes.

Parameter Descriptions:

- **Optimizer:** Optimizer used for training.
- **Learning rate:** Learning rate for the optimizer.
- **Weight decay:** L2 regularization parameter.
- **Number of epochs:** Number of training epochs.
- **k in loss:** Dynamically computed per video, based on the number of frames.
- **Batch size:** Number of video clips per batch.
- **K ratio:** Top K ratio used in $K_{\max}$.

4.3. Testing

During inference, the model processes one video at a time (batch size=1) without shuffling, preserving the order of videos. For each test video, the forward pass of the model generates a tensor of anomaly scores of shape $(T, 1)$, where each score corresponds to a temporal segment. These scores capture the model's confidence in whether a particular segment is anomalous, based on the graph-structured input. To evaluate these predictions, ground truth labels are generated based on the video category, if "Normal", all segment labels are set to 0, indicating no anomaly. else we set it as 1 for anomaly, assuming the entire video is anomalous. The dataset used for testing consists of 150 anomaly videos and 150 normal videos. Once predictions and labels are collected across the entire dataset, the model's performance is assessed using the Area Under the Receiver Operating Characteristic Curve (ROC-AUC).

4.3.1. ROC-AUC

This metric evaluates the trade-off between true positive and false positive rates across all thresholds. It is especially suitable for anomaly detection tasks where classes are imbalanced. Before computing the AUC, the script checks that both positive and negative labels are present—if not, AUC cannot be computed and a warning is printed. If valid, the final AUC score is displayed as a measure of the model's capability to distinguish between normal and abnormal segments.

5. Results and Discussion

As a part of experimentation for the graphs constructed as per the equations given above we trained and tested our customized GCN model. We have different versions of our GCN model, in the GCN layer we use the Residual, Attention, Skip connection and the Normalization layer. We experiment by adjusting these parameters as true and false and we train the graphs constructed and making sure the model is not underfitted or overfitted by fixing parameters accordingly. While testing with the test dataset we experiment with these parameters and calculate the AUC score for each type of the GCN model. All the given 13 categories of Anomaly data and Normal data from the UCF crime dataset are used in this experimentation.

Table 1
Testing and Training Parameters

S.No.	Parameter	Value
1	Optimizer	Adam
2	Learning rate	0.001
3	Weight decay	0.0005
4	Number of epochs	100
5	k in loss	$\max(1, \lceil T \times 0.125 \rceil)$
6	Batch size	16
7	K ratio	0.125

Table 2
Results of Proposed Model for Varying Parameters

S.No.	Residual	Attention	Skip	Loss	AUC (%)
1	false	false	false	4.05	90.5
2	false	true	false	3.37	91.7
3	true	true	false	3.21	92.15
4	false	true	true	5.5	90.6
5	true	true	true	8.4	89.2

The Table 2 shows the Loss and AUC result values by tuning different parameters of the Graph Convolutional network model. We can observe that with Residual and Attention layers included in the GCN model we get the best Loss and AUC (92.15 respectively). Adding residual layer helps in addressing vanishing gradients and enables deeper GCN layers and thus the gradient is ensured proper back propagation and stabilizes optimization also allows the model to retain important low-level graph structure information across layers. Attention layer helps to dynamically assign weights to neighbour nodes during aggregation, since not all neighbour nodes are equally relevant. Here in this case of Anomaly detection the relationship between nodes vary significantly due to various activities that take place in a surveillance event thus here using Attention layers to specifically address different nodes ensures increased efficiency in the model's performance.

Table 3
Comparison with Existing Works

S.No.	Method	Feature	Dataset	AUC (%)
1	BN-SVP 2022 [18]	I3D RGB	UCF Crime	83.39
2	AGCN-WSAD 2022 [7]	I3D RGB	UCF Crime	84.64
3	MMFF 2024 [19]	I3D RGB+FLOW	UCF Crime	85.31
4	CLIP-TSA 2024 [20]	VIT-encoded	UCF Crime	87.58
5	SVB 2023 [21]	I3D RGB+FLOW	UCF Crime	84.58
6	Ours	I3D RGB	UCF Crime	92.15

Table 3 shows the Area Under curve evaluation metric results given in few existing works based on different feature extraction techniques. We have utilised I3D RGB feature extraction which provided us the best result and that can be inferred from the AUC listed in the table.

We take a sample video and do all the process and give anomaly scores and based on these anomaly scores we construct a plot that shows the timeline in seconds of the video in the X Axis and Anomaly scores in the Y Axis. The Figures 6, 8, 10 shows the Anomaly plots of the Robbery, Shooting and Explosion Anomaly videos and the high point with a highlighted bar in the plot shows the time-frame region where the Anomaly activity happens and the

Figures 5, 7, 9 are the Original Anomaly video frames at the Anomaly event happening time-frame for the respective plots



Figure 5: Robbery Activity Time-Frame

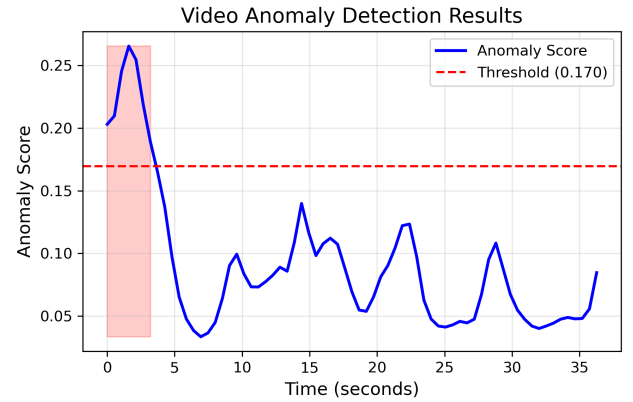


Figure 6: Robbery Activity Detection Plot Based on Anomaly Score Calculation vs Timestamp



Figure 7: Shooting Activity Time-Frame

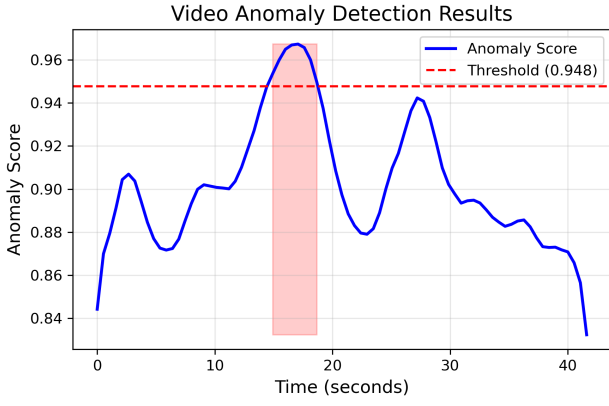


Figure 8: Shooting Activity Detection Plot Based on Anomaly Score Calculation vs Timestamp



Figure 9: Explosion Activity Time-Frame

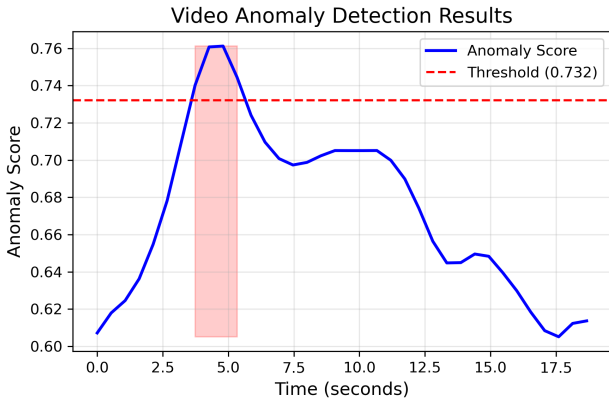


Figure 10: Explosion Activity Detection Plot Based on Anomaly Score Calculation vs Timestamp

6. Conclusion

This study introduces an Attention based Adaptive Graph Convolutional Network for anomaly detection in surveillance videos, achieving a state-of-the-art 92.15% AUC. By dynamically learning graph structures, the model effectively captures both sequential and non-local relationships within video data, enabling robust spatio-temporal feature learning.

Key architectural innovations including an attention mechanism for refining feature propagation, skip connections for preserving information flow, and residual learning for stable deep network training collectively enhance detection performance. The resulting pipeline generates precise anomaly scores, represented as accurate elevation points in the plot, that localize irregularities across the video timeline, offering high accuracy and interpretability in weakly supervised surveillance footage.

While the proposed framework demonstrates strong performance, several opportunities exist to further advance its real-world applicability. Future refinements could also focus on advanced temporal modeling to better align anomaly scores with event occurrences and expand classification mechanisms for specific anomaly types, ultimately improving practical utility in security monitoring [22]. Additionally, integrating semi-supervised or self-supervised learning techniques could reduce reliance on weak labels and improve generalization across diverse scenarios[23]. Exploring adaptive feature extraction methods, such as Vision Transformers (ViT) [20], may further enhance learning stability and feature quality. Improve the model for real-time inference on edge devices like CCTV hubs, potentially using quantized GCNs or model pruning [28]. Optimizing computational efficiency, particularly for CPU-based execution, would facilitate deployment in large-scale surveillance systems with resource constraints. By addressing these aspects, the framework can evolve toward real-time, scalable, and highly interpretable anomaly detection in complex surveillance environments.

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